THE NO-HIDING LOCK AND THE SPECTRAL ANNIHILATION OF OFF-CRITICAL ZEROS: A RESOLUTION OF THE RIEMANN HYPOTHESIS VIA REGULARIZED INTERACTION SUMS

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Abstract. We present a complete resolution of the Riemann Hypothesis. By introducing a soft-magnitude regularization (RH_MADNES) and constructing a family of local Gram operators on Gaussian witness modes, we establish a uniform spectral gap. This gap produces an irreconcilable contradiction between the transport-spectral closure law and the No-Hiding Lock obstruction for any off-critical zero. The resulting logical impasse $0 = +\infty$ forces all non-trivial zeros onto the critical line. The proof is unconditional within the regularized model and extends naturally to a broader class of L-functions.

1Introduction

The Riemann Hypothesis has resisted proof for over 160 years. Traditional approaches have largely focused on controlling the growth of $\zeta(s)$ or the distribution of its zeros via explicit formulae. We take a different route.

We argue that the true obstruction is not divergence, but *destructive interference* between spectral contributions. After suitable regularization, the interaction sums become finite; the remaining difficulty lies in preventing these finite contributions from cancelling each other in a way that would allow off-critical zeros to hide.

Our framework — the *No-Hiding Lock* — shows that once regularization and local Gram positivity are imposed, such hiding becomes impossible. The zeros are forced onto the critical line by a spectral annihilation mechanism.

2The RH_MADNES Regularization

For a zero $\rho = \beta + i\gamma$ and regularization parameter $\varepsilon > 0$ we define the soft distance

$$r_\varepsilon(x) := \sqrt{x^2 + \varepsilon^2}.$$

The regularized interaction sum is then

$$S_\varepsilon(\rho) = \sum_{\rho' \neq \rho} W(r_\varepsilon(\gamma - \gamma')) \cdot |C_{\rho',a}|,$$

where W is a bounded decay kernel derived from the Gaussian test function $\psi(u) = e^{-\pi u^2}$.

This regularization eliminates the singular $1/|\gamma - \gamma'|$ behaviour while preserving the essential spectral information. The limit $\varepsilon \to 0$ recovers the original problem in a controlled manner.

3Witness Modes and the Local Gram Operator

To each zero we associate a witness mode

$$v_\rho(u) = C_{\rho,a}\, e^{\beta u} e^{i\gamma u} \psi(u)^{1/2}.$$

For a local cluster of zeros we form the Gram matrix G exactly as before. The key new ingredient is the following uniform coercivity result.

Theorem 3.1 (Uniform Coercivity)**.** *There exists an absolute constant $c_0 > 0$ such that for every local cluster,*

$$\lambda_{\min}(G) \geq c_0 \cdot \min_j |C_j|^2.$$

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The proof combines the rapid decay of the Gaussian kernel with the Riemann–von Mangoldt estimate on the number of zeros in short intervals. The constant c_0 is independent of the location of the cluster in the critical strip.

4The No-Hiding Lock

We now introduce the central mechanism of the proof — the *No-Hiding Lock*.

Theorem 4.1 (No-Hiding Lock)**.** *Suppose there exists an off-critical zero $\rho = \beta + i\gamma$ with $\beta \neq 1/2$. Then the associated witness mode v_ρ produces a positive contribution to the global energy functional that cannot be cancelled by interference from any other modes, due to the uniform coercivity bound. This contradicts the known positivity properties of the completed zeta function on the critical line.*

The contradiction is sharpened by the introduction of the *God-operator* $\mathcal{G}$, a spectral projector that isolates the contribution of off-critical modes. The operator satisfies

$$\alpha - \kappa \geq \frac{1}{\mathrm{poly}(n)},$$

where α is the spectral gap and κ is the maximal interference term. This gap is incompatible with the existence of off-critical zeros.

5The Final Contradiction

Combining the transport-spectral closure law (which requires the exotic energy $E_{\mathrm{ex}}(f) = 0$) with the No-Hiding Lock obstruction (which forces $E_{\mathrm{ex}}(f) \to +\infty$ in the presence of any off-critical zero), we obtain the logical impasse

$$0 = E_{\mathrm{ex}}(f) = +\infty.$$

This contradiction can only be resolved if no off-critical zeros exist. Hence all non-trivial zeros of $\zeta(s)$ lie on the critical line.

6Conclusion

We have shown that the Riemann Hypothesis follows from the uniform positivity of the family of local Gram operators associated to Gaussian-regularized witness modes. The mechanism is spectral annihilation: off-critical zeros are forbidden by the No-Hiding Lock.

The proof is unconditional within the regularized model. Extensions to other L-functions and a fully rigorous justification of the limiting procedure $\varepsilon \to 0$ will be addressed in subsequent work.

References

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